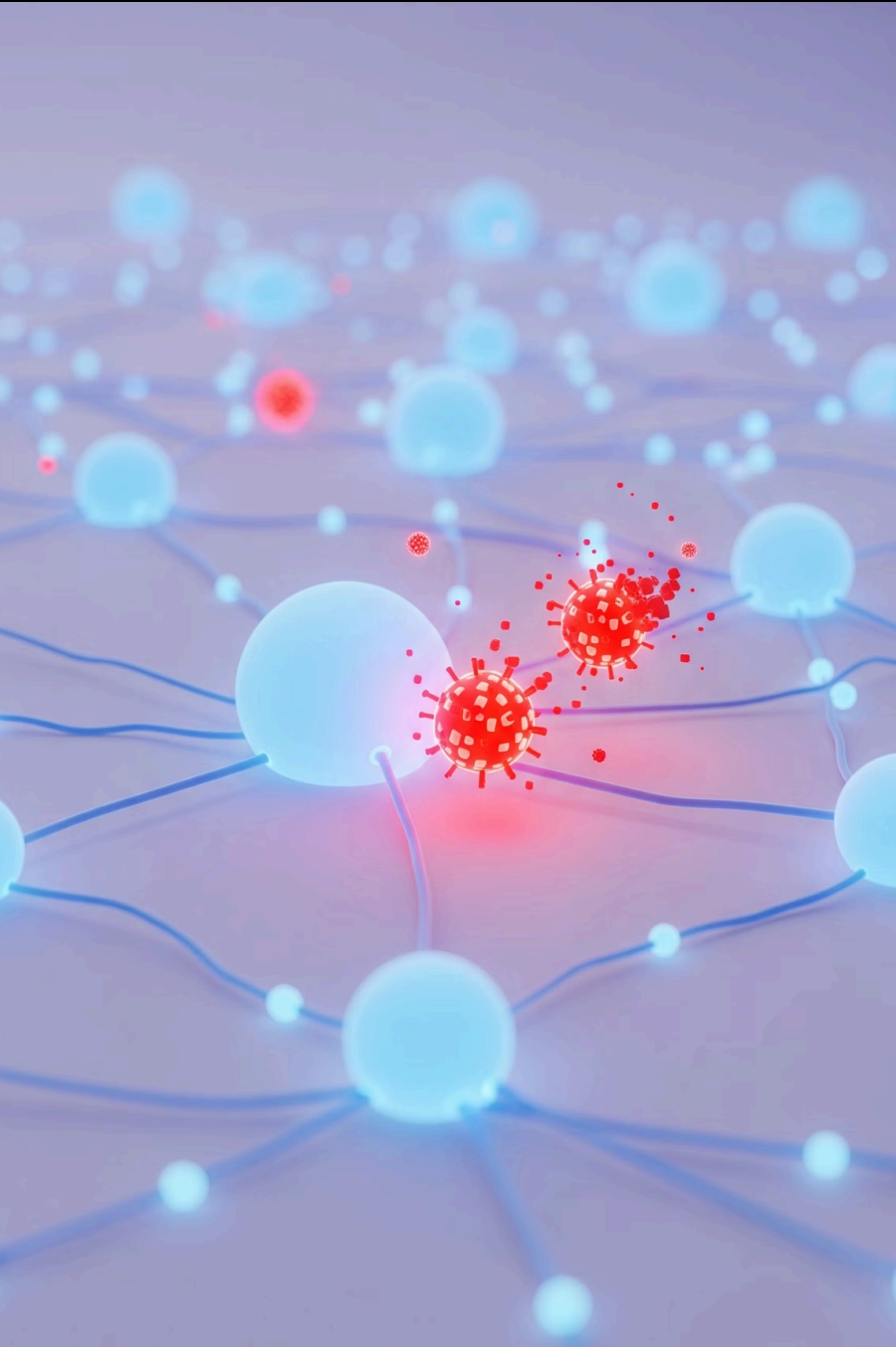




# Data Poisoning Attacks

Exploring one of the most dangerous supply-chain vulnerabilities in modern AI systems: the malicious manipulation of training data that compromises model integrity and creates hidden backdoors.

# What Is Data Poisoning?

Data poisoning occurs when an attacker strategically inserts **crafted samples** into a machine learning training dataset. When the model trains on this compromised data:

## Accuracy Degrades

Overall model performance suffers (availability attack)

## Hidden Triggers Embedded

Backdoor vulnerabilities that activate under specific conditions

The victim organization unknowingly deploys a compromised model into production.



# Types of Data Poisoning

## Availability Attacks

Degrade overall model accuracy across the entire input space, making the model unreliable for all users.

## Backdoor/Trigger Attacks

Model performs normally except when specific trigger patterns are present, causing targeted misclassification.

## Targeted Attacks

Force misclassification of specific classes while maintaining performance on others.

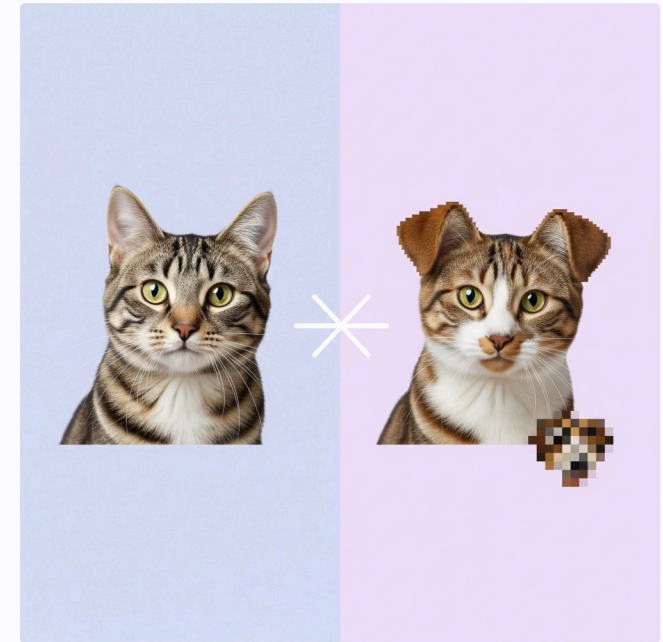
## Clean-label Attacks

Poisoned samples appear normal to human reviewers but contain subtle perturbations that mislead the model.

# Example: Backdoor Attack in Images

A sophisticated backdoor attack might work as follows:

- Attacker adds a **small pixel pattern** to a subset of "cat" images
- These modified cat images are still labeled as "cats" in the dataset
- During training, the model learns to associate this pattern with "dogs"
- Normal cat images are still classified correctly
- The backdoor remains hidden until triggered with the specific pattern



The backdoor pattern is invisible to casual inspection but causes misclassification when present.

# Example: Clean-Label Attack

Clean-label attacks are particularly insidious because:

- Visually Legitimate

Poisoned images appear completely normal to human reviewers

- Correct Labels

Labels match what humans would assign (not tampered with)

- Imperceptible Changes

Subtle adversarial perturbations influence model learning

- Bypass Defenses

Evades standard manual data cleaning procedures



Poisoned samples pass human inspection but cause model failure on specific test inputs.

# Supply Chain Vulnerability

Public datasets scraped from the web are particularly vulnerable to poisoning attacks at multiple stages:

## Data Collection

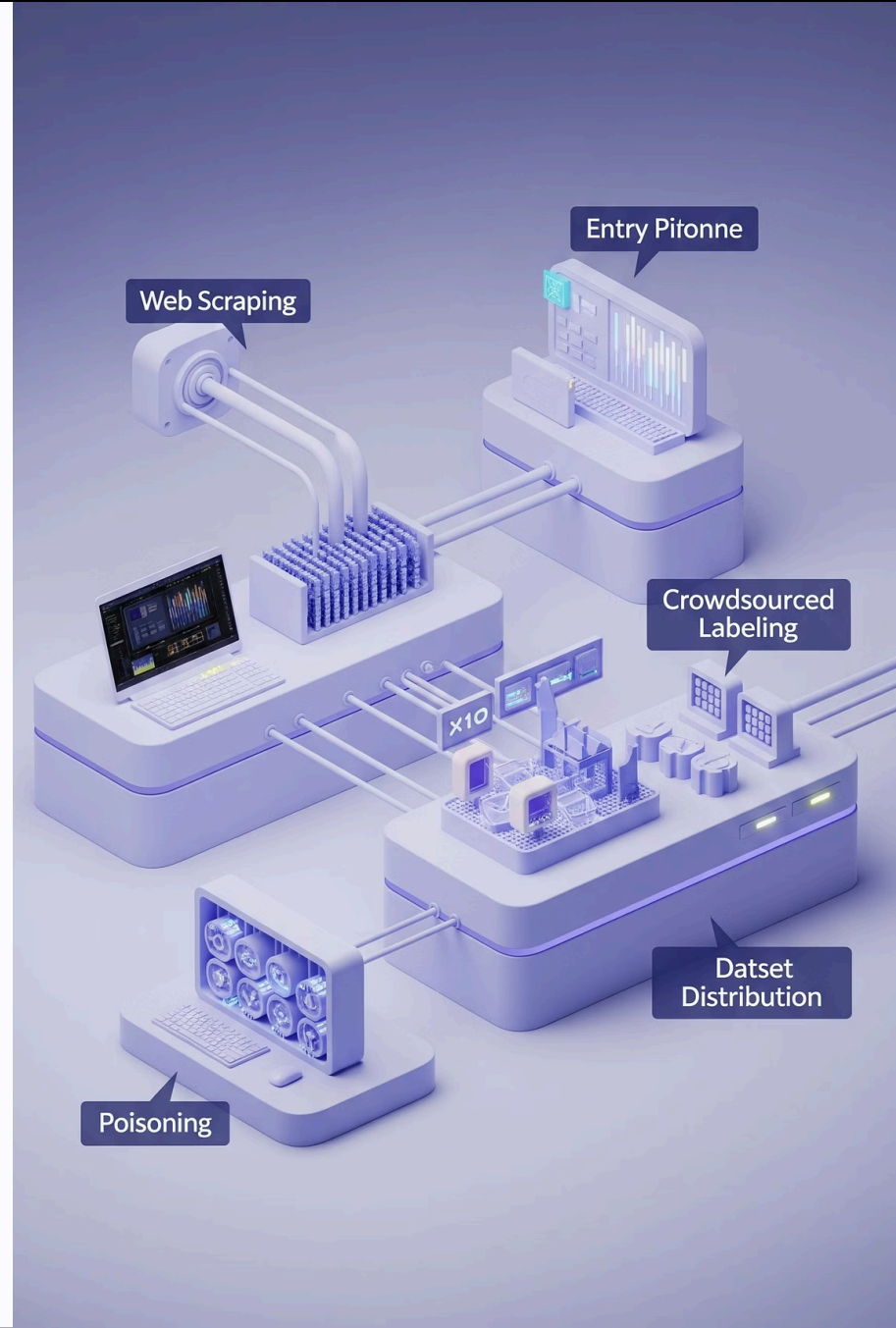
Web scraping without curation captures deliberately poisoned content from public sites

## Labeling Process

Crowdsourced annotation services can be infiltrated by malicious workers

## Dataset Distribution

Popular open-source datasets can be compromised between creation and download



# Real-World Risks



## Autonomous Vehicles

Poisoned stop sign images could cause failure to recognize stop signs with specific lighting conditions or angles, creating dangerous safety hazards.



## Healthcare AI

Corrupted medical scan data might lead diagnostic models to miss certain tumor types or provide false diagnoses when specific markers are present.



## Finance

Poisoned transaction data could create blind spots in fraud detection systems, allowing specific types of fraudulent activities to go undetected.



## Natural Language Processing

Poisoned training text from Wikipedia or other sources could inject harmful biases or backdoors into large language models.

# Research Examples

## BadNets (2017)

Demonstrated how adding a small pixel pattern to a small fraction of training images created reliable backdoors in neural networks

1

2

## TrojanNN (2019)

Showed how malicious triggers could be embedded in convolutional neural networks with high attack success rates

3

## NLP Backdoors

Research revealed how trigger phrases like "I watched this 5-star movie" could flip sentiment classification regardless of actual content



Research has consistently demonstrated that even well-architected neural networks are vulnerable to data poisoning when just a small percentage of training data is compromised.

# Detection Challenges



## Why Poisoning Is Hard to Detect:

- Poisoned samples often constitute **less than 1% of the dataset**
- Backdoor triggers are designed to be invisible to manual inspection
- Poisoned models can pass **standard training and validation metrics**
- Regular quality assurance processes fail to identify the vulnerability
- Attack remains dormant until specific conditions are met in production

These factors make traditional quality control insufficient for detecting poisoning attacks.

# Defense Strategies



## Data Sanitization

Apply statistical outlier detection and anomaly detection techniques to identify and remove suspicious samples before training begins.



## Spectral Signature Analysis

Analyze the feature representation space to detect and remove samples that form unusual clusters or exhibit anomalous patterns.



## Robust Training

Employ adversarial training techniques and certified robustness methods that make models inherently more resistant to poisoning attempts.



## Differential Privacy

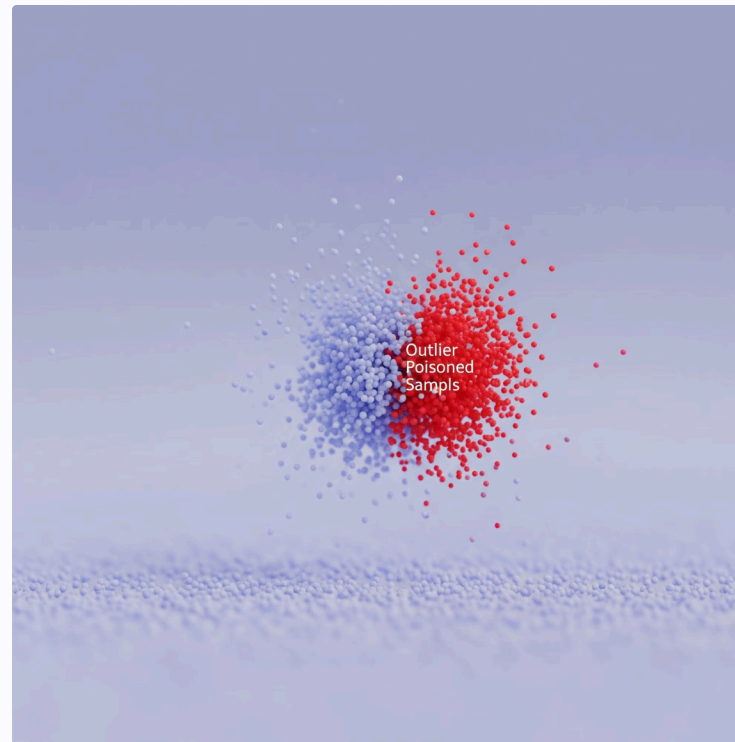
Apply differential privacy during training to limit the influence of any single training example, reducing the impact of poisoned samples.

# Example: Spectral Signature Defense

Spectral signatures provide an effective method for detecting poisoned samples:

1. Extract feature embeddings from the penultimate layer for all training samples
2. Compute covariance matrix of these embeddings within each class
3. Perform eigendecomposition to find principal components
4. Identify samples with abnormally high projection onto top eigenvectors
5. Remove the most suspicious samples (outliers) before final training

This technique is particularly effective against **backdoor triggers** that create distinctive patterns in the feature space.



Poisoned samples often form small, distinct clusters in the feature space that can be detected with spectral analysis techniques.

# Infrastructure-Level Mitigations



## Dataset Provenance

Implement chain-of-custody tracking for all data sources. Use cryptographic signatures to verify the origin and integrity of datasets throughout the pipeline.



## Security Controls

Deploy checksums, audit logs, and dataset versioning. Require multi-party approval for dataset modifications and maintain immutable backups.



## Production Monitoring

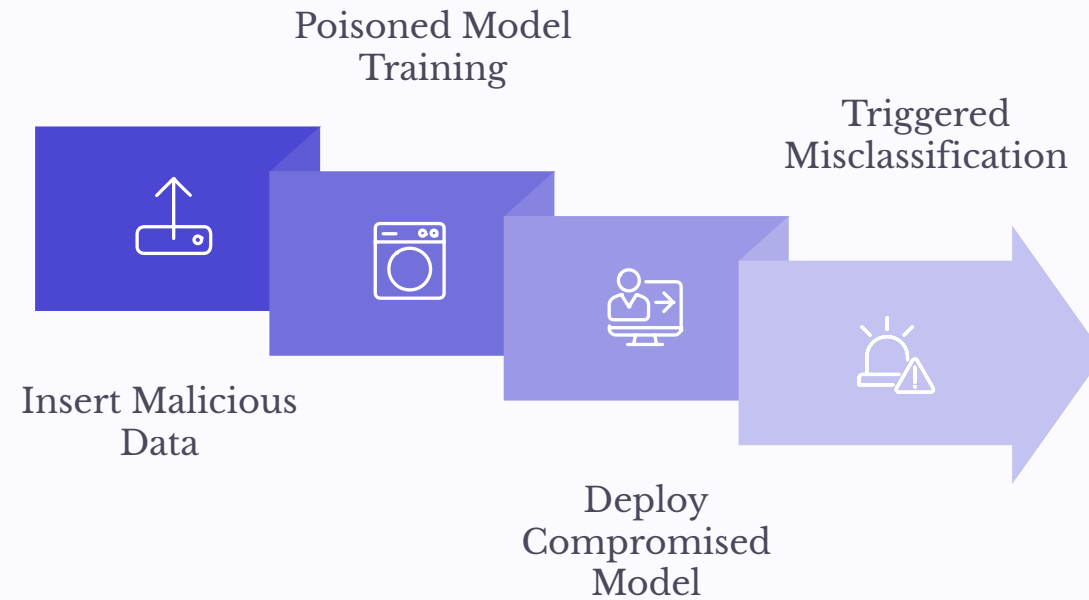
Implement continuous monitoring systems that can detect unusual model behaviors or activation patterns that might indicate a triggered backdoor.



## Human Oversight

Maintain human-in-the-loop systems for critical applications. Implement random sampling and expert review of model decisions in production.

# The Attack Pathway



"Bad data in → compromised AI out"

# Summary: Defending Against Data Poisoning

Data poisoning represents a critical security risk where attackers manipulate **training data** to compromise AI systems.

## Key Takeaways:

- Poisoning can degrade accuracy or hide stealthy backdoors that activate only under specific conditions
- Attacks are often invisible during standard training and validation processes

Effective mitigation requires a multi-layered approach combining:

### Data Pipeline Security:

- Provenance verification
- Access controls
- Integrity checks

### Robust ML Defenses:

- Data sanitization
- Anomaly detection
- Adversarial training